ConsentChain: Contract-Enforced Dynamic Consent and Verifiable Federated Unlearning for IoT on PureChain

Josiah Ayoola Isong, Simeon Okechukwu Ajakwe (SMIEEE)*, Jae-Min Lee, Dong-Seong Kim (SMIEEE)
Department of IT Convergence Engineering, Kumoh National Institute of Technology, Gumi, South Korea
*Smart Computing Department, Kyungdong University, Global Campus, South Korea
isongjosiah@gmail.com, profsimeon@kduniv.ac.kr, (ljmpaul, dskim)@kumoh.ac.kr

***Abstract*—Federated learning over consumer IoT must now honour data-subject withdrawal under GDPR, the EU AI Act, and CCPA: consent can be revoked and the trained model must demonstrably forget. Prior work treats consent management, machine unlearning, and on-chain auditability as disjoint threads. ConsentChain closes this *consent–unlearning–verification triangle* in one chain-enforced pipeline: a Consent Lifecycle State Machine as soulbound ERC-721 tokens, a Verifiable Unlearning Protocol gating on-chain acceptance on three independent membership-inference attacks, and a three-contract Solidity architecture on gas-free PureChain (PoA^2) and gas-based Amoy (PoS). Across N-BaIoT and CASAS, per-subject lifecycle cost is $0 on PureChain versus $0.00120 on Amoy, and the gate accepts approximate unlearning (MIA < 0.55) yet rejects naive retraining (whose distributional uniqueness inflates MIA) and any attestation lacking a 3-of-5 verifier threshold, staying faithful to MIA evidence rather than the controller's claim.**

***Index Terms*—federated learning, machine unlearning, smart contracts, dynamic consent, edge intelligence, trustworthy AI, IoT, GDPR, EU AI Act, membership inference attack**

I. INTRODUCTION

Federated learning (FL) trains a shared model without centralising client data [1], often the only legally viable option in consumer IoT [2]. But GDPR Articles 7 and 17 demand purpose-granular, revocable consent whose withdrawal propagates into the model. Federated unlearning (FUL) erases a client's data and influence, yet a bare withdrawal is a one-shot trigger, not a scoped state binding round selection, so training on revoked data may continue unrecorded; and approximate unlearning, the only kind tractable at FL scale, offers no guarantee, fails *silently*, and is self-reported by the controller that owes it. ConsentChain couples *dynamic consent* with *verifiable machine unlearning* in one chain-enforced protocol, closing both gaps.

ConsentChain makes four contributions: (1) a Consent Lifecycle State Machine (CLSM), a soulbound-ERC-721 automaton of six states and nine transitions whose safety and liveness hold by construction; (2) a Verifiable Unlearning Protocol (VUP), method-agnostic, accepting a UA iff three *independent* MIAs (shadow [3], LiRA [4], label-only [5]) fall below τ under t-of-n ECDSA signatures; (3) three Solidity contracts (ConsentRegistry, UnlearningOracle, FLGatekeeper) on both a gas-free (PureChain, PoA^2) and gas-based (Amoy, PoS) chain, the first side-by-side cost comparison; and (4) a dual-domain evaluation (9-client N-BaIoT [6] autoencoder, 5-client CASAS BiLSTM) over three approximate-unlearning methods plus exact retraining, measuring forgetting, cost, and utility recovery.

II. RELATED WORK

Three threads each solve part of the problem but none their conjunction. MedRec [7] records consent immutably yet treats the record as an endpoint: no training is gated by it. Federated-unlearning methods, SISA [8] (shard retraining), FedEraser [9] (history replay), and gradient ascent, remove a contribution but, like the broader survey [10], never specify how unlearning is triggered or bound back to consent. Blockchain-FL brings computation on-chain without consent: BlockFL [11] and BFLC [12] log updates yet let no participant withdraw, and the closest work, Zuo et al. [13], certifies that an unlearning routine *ran*, not that the model *forgot*, the very gap membership inference [3]–[5] measures (a single attack can overstate it [14], motivating our three-attack conjunction).

TABLE I
PRIOR WORK ACROSS THE FOUR CONTRIBUTION DIMENSIONS PLUS COST. *Consent* = GATES TRAINING (NOT MERELY RECORDED); *Verify* = MEASURES FORGETTING (NOT MERELY RECORDS THE ACTION); *Partial* = RECORDED BUT NOT ENFORCED OR MEASURED.

System	Consent	Unlearn	Verify	On-chain	Gas-free
MedRec [7]	Partial	No	No	Partial	No
SISA/FedEraser [8], [9]	No	Yes	No	No	n/a
BlockFL/BFLC [11], [12]	No	No	No	Yes	No
Zuo et al. [13]	No	Yes	Partial	Yes	No
ConsentChain (ours)	Yes	Yes	Yes	Yes	Yes

Each prior system covers at most one dimension; ConsentChain supplies the missing edges (Table I): consent state gating rounds under CLSM properties, any unlearning paradigm as a black box, and three-MIA verification driving the consent transition. Chain choice is economic, not protocol: identical contracts run on PureChain [15], a public gas-free PoA^2 chain, and on gas-based Amoy, a difference quantified in Section IV.

III. SYSTEM MODEL AND FRAMEWORK

A. System model

Six entities $\langle DS_i, DC, \{FL_k\}, VC, CR, UO, GK\rangle$ compose ConsentChain: data subject DS_i (key-controlled owner of one soulbound Consent NFT), controller DC (drives

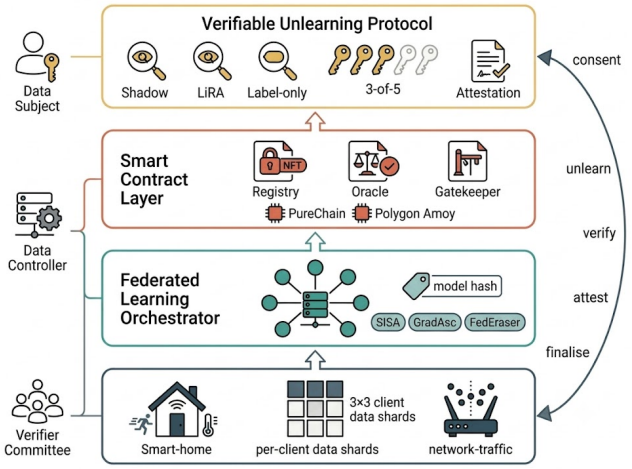


Fig. 1. ConsentChain architecture: off-chain IoT data and Flower FedAvg layers, the three on-chain Solidity contracts on PureChain / Polygon Amoy, and the off-chain Verifiable Unlearning Protocol. The clockwise loop consent→unlearn→verify →attest→finalise is the edge ConsentChain closes.

the round loop, triggers unlearning on revocation), devices FL_k (local SGD), Verifier Committee $VC = \{v_1, \ldots, v_n\}$ ($n = 5$ ECDSA keys, threshold $t = 3$), and the three contracts `ConsentRegistry` (CR), `UnlearningOracle` (UO), `FLGatekeeper` (GK); any chain reader is an *auditor*. ConsentChain runs FedAvg [1],

$$\theta^{(r+1)} = \sum_{k \in K_r} \frac{n_k}{\sum_{j \in K_r} n_j} \theta_k^{(r+1)}, \tag{1}$$

with the participant set K_r in (1) fixed to GK.`validateRound`'s return value, not an orchestrator-chosen subset: consent is enforced at the membership, not the gradient, level.

B. Consent Lifecycle State Machine

The CLSM is a deterministic finite automaton CLSM $= (S, \Sigma, \delta, s_0, F)$ with start $s_0 = \textsc{Ina}$, absorbing terminal $F = \{\textsc{Unl}\}$, and states $S = \{\textsc{Ina}, \textsc{Act}, \textsc{Sus}, \textsc{Rev}, \textsc{Exp}, \textsc{Unl}\}$ (inactive, active, suspended, revoked, expired, unlearned). Its nine δ-edges over the eight events Σ are

$$\begin{aligned} &\texttt{grant}: \textsc{Ina} \to \textsc{Act}, \quad \{\texttt{modify}, \texttt{tick}\}: \textsc{Act}\circlearrowleft, \\ &\texttt{revoke}: \textsc{Act} \to \textsc{Rev}, \quad \texttt{expire}: \textsc{Act} \to \textsc{Exp}, \\ &\texttt{suspend/resume}: \textsc{Act} \leftrightarrow \textsc{Sus}, \\ &\texttt{unlearn_verified}: \{\textsc{Rev}, \textsc{Exp}\} \to \textsc{Unl}. \end{aligned} \tag{2}$$

Safety is immediate from (2): no edge targets ACT from $\{\textsc{Rev}, \textsc{Exp}\}$, `unlearn_verified` is oracle-only, and the lone `validateRound` edge `tick` guards on $\rho_r \in \mathcal{P}$. `ConsentRegistry` realises δ one-edge-per-function; the soulbound constraint (no `transferFrom`) confines token-i transitions to DS_i.

C. Verifiable Unlearning Protocol

On $\text{state}(i) \to \{\textsc{Rev}, \textsc{Exp}\}$ the gatekeeper sets `pendingUnlearningSince`$[i] = b$ and `validateRound` blocks all rounds (liveness). Within $\Delta_{\max}$ blocks the controller must unlearn, score f_{post} with the three MIAs, assemble the Unlearning Attestation, gather t-of-n signatures, and submit.

Unlearning methods. ConsentChain is method-agnostic over the three families of Section II: SISA-Fed [8] (per-shard retrain), gradient ascent ($\theta^{(j+1)} = \theta^{(j)} + \eta \nabla \mathcal{L}(\theta^{(j)}; \mathcal{D}_i)$, $k = 10$ steps, $\eta = 10^{-4}$), and FedEraser [9] (history replay with 3-round calibration).

Membership-inference attacks. Against DS_i's set $\mathcal{D}_i$ (members M), three independent attacks each score f_{post} in $[0, 1]$ (higher = more residual membership). The *shadow-model* attack [3], adapted to reconstruction models [16], trains an MLP classifier on the "in"/"out" reconstruction errors of $S = 5$ shadow autoencoders; s_{sh} = the fraction of M it flags "in." *LiRA* [4], the strongest, fits in-/out-pool Gaussians to the log-reconstruction-errors of $n_{\text{in}} = n_{\text{out}} = 4$ reference models per $x \in M$ and takes s_{lira} = TPR at an α-controlled FPR ($\alpha = 0.05$; $\alpha = 0.5$ ablation in Section V). The *label-only* attack [5], seeing only the reconstruction decision, thresholds at FPR 0.05 and takes s_{lo} = the fraction of M below it.

Attestation and hash binding. The Unlearning Attestation is the tuple $u = \langle i, h_{\text{pre}}, h_{\text{post}}, m, s_{\text{sh}}^{\text{fp}}, s_{\text{lira}}^{\text{fp}}, s_{\text{lo}}^{\text{fp}}, t_u \rangle$: token ID, pre-/post-unlearning model hashes, method $m \in \{0, 1, 2\}$, fixed-point scores $s_j^{\text{fp}} = \lfloor 10^4 s_j \rfloor$, timestamp. Each verifier signs $h_u = \text{keccak256}(c \,\|\, u)$ over $c =$ `block.chainid`, binding the UA to one chain against cross-chain replay. Writing $R(\Sigma)$ for the distinct registered signers of h_u and $h_{\text{rev}}(i) =$ CR.`modelHashAtRevocation`(i), the oracle finalises iff

$$\begin{aligned} \text{accept}(u, \Sigma) : \Big(\textstyle\bigwedge_j s_j^{\text{fp}} < 5500 \Big) \;\wedge\; |R(\Sigma)| \geq t \\ \wedge\; h_{\text{pre}} = h_{\text{rev}}(i) \;\wedge\; \text{state}(i) \in \{\textsc{Rev}, \textsc{Exp}\}. \end{aligned} \tag{3}$$

The last two conjuncts pin the UA to the exact model version and block re-attestation of an already-UNL token, closing the replay loop. With $\tau^{\text{fp}} = 5500$ just above the 0.5 random-guess baseline (margin for finite-$|M|$ noise), the three-attack conjunction is what makes so tight a threshold tolerable.

D. Three-contract Solidity architecture

ConsentChain is three Solidity 0.8.24 contracts (NatSpec in supplementary `docs/contracts-spec.md`). `ConsentRegistry`, the soulbound ERC-721 realising δ above, adds a write-once h_{rev} and a `transitionToUnlearned` callable only by the oracle. `UnlearningOracle` holds VC, t, and τ^{fp}; `submitAttestation` evaluates $\text{accept}(u, \Sigma)$ and, on success, atomically finalises the token and unblocks the gatekeeper. `FLGatekeeper` gates rounds: `validateRound` reverts on any ineligible participant or pending unlearning (encoding liveness), and permissionless `reportTimeout` emits `ComplianceViolation` past the $\Delta_{\max}$ deadline. The loop closes when a revocation fires `recordPendingUnlearning` and only a valid UA, via the oracle's `unblock`, releases the gatekeeper (Figure 1).

E. Threat model

We defend four in-scope adversaries: a *lazy orchestrator* (GK reverts on pending unlearning, emits `ComplianceViolation` after $\Delta_{\max}$), a *deceptive controller* (three MIAs plus VC t-of-n signatures), a *consent forger* (soulbound NFT, `dsOwner` check), and a *replay attacker* (UA binding (c, i, h_{pre})). The committee is the trust floor: $\geq t$ colluding verifiers can certify an invalid attestation. Our journal remedy stakes verifiers, slashable on a successful challenge: DS_i, the one party holding $\mathcal{D}_i$ and motivated to dispute, contests an accepted UA by exhibiting a contradicting MIA evaluation against the attested $(h_{\text{pre}}, h_{\text{post}})$, with per-attestation committees VRF-drawn from a larger pool against pre-collusion. Colluding participants (an orthogonal DP layer) and six further classes are out of scope (Section V); the full per-adversary model is in supplementary `docs/threat-model.md`.

F. Formal properties

ConsentChain satisfies four properties: the first pair by construction, the second empirically (Section IV). *Safety (consent integrity):* `validateRound` admits only ACT tokens with $\rho_r \in \mathcal{P}_k$, and no δ-path re-enters ACT (above). *Audit completeness:* every transition, attestation, violation, and model-hash update emits an indexed event. *Enforceability (liveness):* within $\Delta_{\max}$ blocks of a revocation, either a valid UA drives the token to UNL or the gatekeeper logs a public `ComplianceViolation`, degrading to "no training," not "unconsented training." *Verifiability:* the oracle finalises iff $\text{accept}(u, \Sigma)$ (3) holds. Mechanised proofs are left to future work.

IV. EVALUATION

We evaluate unlearning effectiveness, on-chain cost (PureChain versus Polygon Amoy), round-validation scalability to $N = 500$, and utility recovery, plus two on-chain demos of the liveness and verifiability properties; the reproducibility recipe is in supplementary `docs/eval-protocol.md`.

A. Experimental setup

The primary federation is a 9-client N-BaIoT anomaly detector [6], one autoencoder per device, a symmetric MLP $115 \to 75 \to 50 \to 25 \to 10$ trained with FedAvg for 30 rounds (Flower, seed 42). A GDPR Art. 17 erasure of client 5 at round 30 triggers one of the three approximate methods or exact 8-client retraining. Contracts run live on gas-free PureChain (PoA^2) and gas-based Amoy (PoS), with VC of five ECDSA keys, $t = 3$, $\Delta_{\max} = 1000$ blocks.

B. Unlearning effectiveness

Table II reports each method's three MIA scores on the revoked client; the `submitAttestation` predicate accepts iff all three sit below $\tau = 0.55$.

The gate accepts all three approximate methods: shadow MIA reads 0.438/0.254/0.523 (gradient ascent/SISA-Fed/FedEraser), with LiRA and label-only far below τ. Strikingly, it *rejects* exact naive retraining, which never trained on client 5: shadow scores $0.584 > \tau$ because it reads client 5's distinctive traffic distribution, not membership [3], a confound LiRA (0.045) and label-only (0.014) sidestep. The no-unlearning baseline already sits near random (0.55/0.05/0.01), so this federation memorises client 5 only weakly: here the gate's role is rejecting that distributional confound, not detecting strong leakage. Section IV-E supplies the high-memorisation regime where it must, and does, reject real residual signal. All three accepted UAs finalise end-to-end on PureChain to UNLEARNED.

TABLE II
UNLEARNING EFFECTIVENESS: MIA SCORES ON THE REVOKED CLIENT'S DATA, POST-UNLEARNING. THE GATE ACCEPTS IFF ALL THREE ARE BELOW $\tau = 0.55$.

Method	Shadow MIA	LiRA	Label-only	On-chain accept
No unlearning (baseline)	0.5455	0.0475	0.0060	Yes
Gradient ascent	0.4380	0.0435	0.0030	Yes
SISA-Fed	0.2542	0.0795	0.0862	Yes
FedEraser	0.5232	0.0325	0.0050	Yes
Naive retrain (exact ref.)	0.5843	0.0445	0.0138	No

C. On-chain cost and scalability

Table III reports per-operation gas and cost from a lifecycle $\times 10$ run (USD at POL/USD $= 0.0953$).

TABLE III
PER-OPERATION GAS AND COST, LIFECYCLE $\times 10$ MEAN. PER-DS LIFECYCLE = SUM OF THE FOUR OPS.

Operation	PureChain gas	Amoy gas	Amoy MATIC	Amoy USD
`updateModelHash`	30,074	32,673	0.0010	\$0.00009
`grantConsent`	144,868	153,568	0.0046	\$0.00044
`revokeConsent`	107,904	96,404	0.0029	\$0.00028
`submitAttestation` (3-of-5)	102,903	137,600	0.0041	\$0.00039
Per-DS lifecycle	**385,750**	**420,245**	**0.0126**	**\$0.00120**

Per-subject lifecycle cost is \$0 on PureChain and \$0.00120 on Amoy; the gap is gas *price* (zero versus $\sim$ 30 gwei), not count. `validateRound`(N) scales exactly linearly: PureChain runs the full $N \in [5, 500]$ sweep live ($R^2 \approx 1$), while Amoy, measured at $N \in \{5, 10\}$ and extrapolated, projects to \$0.0176 per $N = 500$ round, i.e. \$1.76/day at 100 rounds/day versus \$0; off-chain, `validateRound` adds only $\approx$3.2 ms per round at $N = 500$. Gas-free infrastructure is thus a different cost class, not an optimisation. Two short PureChain runs confirm the remaining properties: a $\Delta_{\max} = 2$ timeout fires `ComplianceViolation` (liveness), and a t-of-5 sweep exercises both branches of the acceptance predicate (verifiability).

D. Utility recovery after revocation

Resuming FL for ten rounds after dropping client 5, the post-revoke model reaches reconstruction error 0.4720, 6.1% *better* than the 9-client baseline (0.5025), since client 5 was an outlier. Gradient ascent (0.4726) and FedEraser (0.4698) match this at no utility cost, while SISA-Fed pays +29% (0.6081 by round 40), consistent with its stronger forgetting (Figure 2).

E. Cross-domain validation: the high-memorisation regime

CASAS provides the complementary high-memorisation regime: where N-BaIoT's baselines sit near random, the 5-client CASAS BiLSTM reaches pre-unlearning shadow MIA

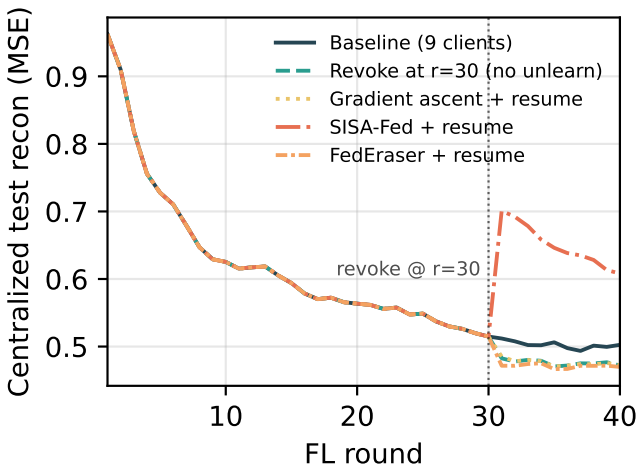


Fig. 2. Utility recovery. Centralized test reconstruction error per FL round. The four post-revocation trajectories diverge at round 30; SISA-Fed pays a real utility cost, gradient ascent and FedEraser converge to within 0.5% of plain post-revoke by round 40.

0.979, so the gate must *reject*, and does: gradient ascent (0.980), FedEraser (0.840), and even exact naive retrain (0.846) stay above τ and are refused, while only SISA-Fed (0.499/0.035/0.000) clears all three attacks. The two domains thus exercise both branches of the acceptance predicate on identical contracts; full results appear in the journal version.

V. Discussion and Conclusion

Our gate is deliberately conservative: shadow MIA conflates distributional uniqueness with membership (naive retrain, never trained on the data, scored 0.584/0.846), so requiring *all three* MIAs below τ errs toward false rejection, and LiRA, which resists the confound but is α-sensitive, is co-calibrated with τ per model class. ConsentChain targets *approximate* unlearning, meeting GDPR Art. 17's "reasonable measures" by predicate plus on-chain audit rather than formal erasure: certified erasure stays intractable for approximate methods at FL scale, so the gate is evidence-based by design. Formal-erasure and ECDSA-leakage guarantees are out of scope (both BLS-collapsed in the journal), as is preventing off-chain training on revoked data *before* a round runs — a controller that does so still cannot produce a passing attestation, so the omission is publicly detectable. Our federations are dataset-bound — nine N-BaIoT device traces, five CASAS apartments — while the on-chain path scales independently to $N = 500$; carrying the learning and MIA side to hundreds of clients is journal work. Its consent, liveness/verifiability, and audit mechanisms discharge GDPR Arts. 7/17/21, EU AI Act Arts. 15/26, and CCPA § 1798.105, and run unchanged on a permissioned-yet-public PoA^2 chain or permissionless Amoy, so the economic model, not the protocol, is the deployment variable.

ConsentChain is, to our knowledge, the first protocol to bind dynamic consent, machine unlearning, and on-chain verification in one chain-enforced pipeline, closing the consent–unlearning–verification triangle prior work addressed only at the vertices. Across 9-client N-BaIoT [6] and 5-client CASAS, approximate unlearning drives all three MIAs below $\tau = 0.55$ at $\leq 3\%$ utility loss, with per-subject lifecycle cost \$0 on PureChain versus \$0.00120 on Amoy. Beyond the BLS aggregation and formal-erasure guarantees noted above, future work targets zero-knowledge eligibility proofs, PoA^2-native verifiers, and cross-chain consent portability.

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